

Generation Challenge

Distribution learning & sampling-time metric optimization on SMILES

Simon Danninger - k12310157

JKU Linz – AI in Life Sciences

1. The challenge

Task:

From a 1,272,851-molecule corpus of drug-like SMILES, train a model to generate 10,000 SMILES.

Scoring:

- ▶ **Fréchet ChemNet Distance (FCD)** Fréchet distance between Gaussian approximations of ChemNet activations of two sets of molecules. Primary metric. Lower is better.
- ▶ Auxiliary metrics:
 - ▶ validity
 - ▶ uniqueness
 - ▶ novelty

must each be ≥ 0.9 or the submission earns no positive grade.

2. EDA on train set

Structure

- ▶ 100% RDKit-valid, 99.93% already canonical
- ▶ Atoms: C, N, O, F, S, Cl, Br, P, I; rare B, Si, Se
- ▶ Charged: 5.8%; stereo: 0%
- ▶ 81% unique Murcko scaffolds (high diversity)

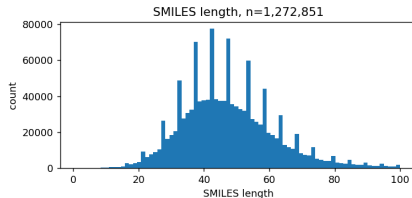


Figure: Length Distribution

Implications for modelling

- ▶ *Char-level* tokenization is wrong: it splits C1 and [nH] → regex SMILES tokenizer
- ▶ hard limit to length 110 for sampling
- ▶ Aux thresholds easy if we post-process (canonicalize → dedup → drop train overlaps → first 10k).

3. Approach

Two-phase strategy.

1. **Distribution learning.** Train an autoregressive SMILES LM on the regex-tokenized corpus. Sample at temperature $T = 1$.
2. **Sampling-time metric optimization.** Generate a large pool of candidates, then *select* the 10k subset whose ChemNet (μ, Σ) best matches the reference set.

Two infrastructure pieces that mattered.

- ▶ **Local FCD harness.** the eval bundle contains `test_stats.p`, which is the *exact* reference (μ, Σ) the server uses. So we can compute the official FCD locally and optimize against it directly.
- ▶ **Aux-metric guard.** Post-process for auxiliary metrics.

4. Model versions

ver.	what	params	train	official FCD
v0	pipeline smoke + local FCD ref	–	–	–
v1	LSTM SMILES LM (regex tokenizer)	3.75 M	5 min	0.758
v2	Transformer LM (6L, dim 384)	10.76 M	25 min	0.359*
v3	Scaled Transformer (8L, dim 512)	25.36 M	80 min	0.258*
v4	subset selection vs <i>local</i> ref (50k pool)	–	1 h	0.208
v5	subset selection vs <i>official</i> ref, 2k swaps	–	0.5 h	0.157*
v6	subset selection, 200k pool, 8k swaps	–	2.5 h	0.083*

* Submitted to the server

v1–v3 = scaling the generator, **v4–v6** = exploiting subset selection.

5. Phase 1: scaling the generator

Versions

- ▶ Regex SMILES tokenizer, vocab = 100
- ▶ v1: 2-layer LSTM, 3 epochs
- ▶ v2: 6-layer pre-norm Transformer, KV-cache sampling
- ▶ v3: 8-layer, dim 512, 5 epochs, cosine LR + AdamW + AMP

ver.	train ppl	FCD (official)
v1	1.95	0.758
v2	1.77	0.359
v3	1.63	0.258

Lessons

- ▶ Temperature sweep is unimodal: $T = 1.0$ wins by a clear margin every time
- ▶ Diminishing returns are visible by v3 ($\sim 10\%$ ppl drop per doubling of params)
- ▶ v3 *natural-sample* FCD is the “honest” generator quality: **0.258**

6. Phase 2: subset selection

Key realization. The eval bundle contains `test_stats.p` = the official $(\mu_{\text{ref}}, \Sigma_{\text{ref}})$. So we can optimize against the grader's metric directly.

Algorithm: swap_full. Given a pool of N pre-generated SMILES with cached ChemNet activations:

1. Start from a random 10k subset.
2. Sample candidate (in, out) pairs; **accept** the first 1-element swap that strictly decreases the full FCD.
3. Repeat until the accepted-swap budget is hit.

Implementation details that mattered

- ▶ Incremental μ/Σ via rank-1 updates of running sum + scatter
- ▶ $O(1)$ in/out bookkeeping with boolean mask

7. Subset selection results

Two findings worth flagging

- ▶ *Pure oversample + random 10k doesn't help*
- ▶ v6 went *below* the full pool's natural FCD (0.083 vs 0.099) – swap actively fits the reference better than the model's own population statistics.
- ▶ Power-law decay: $\text{FCD} \approx 8.3 \cdot n^{-0.51} \rightarrow$ no plateau in 8k swaps.

One wrong method. `greedy_mean` (only minimize $\|\mu_{\text{sub}} - \mu_{\text{ref}}\|^2$) overfitted mean $\rightarrow$ worse than full fit FCD

step	FCD (official)
v3 natural 10k sample	0.258
v4 swap, 50k pool,	0.208
v5 swap, 50k pool, 2k	0.157
v6 swap, 200k pool, 8k	0.083
(full 200k pool, no swap)	0.099

8. Final results

Server submissions.

#	file	aux (v/u/n)	official FCD
1	v2_T1.0	1.0 / 1.0 / 1.0	0.359
2	v3_T1.0	1.0 / 1.0 / 1.0	0.258
3	v5 swap	1.0 / 1.0 / 1.0	0.157
4	v6 swap	1.0 / 1.0 / 1.0	0.083

Honest reading of the numbers.

- ▶ The generator's *natural* quality is v3 at **0.258** (for a 10k sample size). That number is what the model alone produces.
- ▶ The drop $0.258 \rightarrow 0.083$ comes entirely from *subset selection against the released `test_stats.p`*. It is *metric optimization*, not improved distribution learning.
- ▶ The $1/\sqrt{n}$ trajectory suggests the metric can be driven essentially arbitrarily low with more swaps

9. Lessons

1. **Read the eval pipeline carefully.** Discovering that the bundle ships `test_stats.p` reframed the whole problem from “model the distribution” to “hit these specific (μ, Σ) at 10k samples”
2. **Optimize the full objective.** `greedy_mean` is cheaper to compute but does not give the full picture.
3. **Distinguish model improvement from metric optimization.** Bigger model lowered the *natural* FCD; subset selection lowered the *reported* FCD.

Thank you.